Electric Vehicle Market Performance & Insights

Project Report
Date: 1 November 2026
Team ID: _____
Platform: Tableau Public

1. INTRODUCTION

1.1 Project Overview

This project presents an exploratory analysis of electric-vehicle market data through a Tableau Public dashboard. The dashboard brings together eight visualizations covering manufacturer sales, annual sales trends, battery technology distribution, driving range, vehicle price, customer ratings, CO■ savings and manufacturing geography.

1.2 Purpose

The purpose is to make the supplied EV dataset easier to explore and interpret by organizing key measures into visual comparisons. The dashboard is intended to help viewers examine differences across manufacturers, vehicle categories, battery types and countries, and to identify patterns that may guide further analysis.

2. IDEATION PHASE

2.1 Problem Statement

EV market information can be difficult to interpret when sales, technology, vehicle range, pricing, customer ratings and environmental measures are considered separately. The project addresses this by bringing these perspectives together in one visual dashboard. The exact user-research evidence and interview records were not supplied; therefore this statement represents the project framing rather than a validated interview finding.

2.2 Empathy Map Canvas

The intended dashboard audience may include students, analysts, EV researchers and people comparing market-level EV information. They need clear comparisons, readable labels and a concise view of different EV indicators. This is a proposed audience profile, not a result from documented customer interviews.

2.3 Brainstorming

The project scope was organized around eight analytical questions: which manufacturers lead annual sales; how sales vary by model year; how battery technologies are distributed; which vehicle categories have the greatest average range; how price relates to range; how customer ratings compare; which manufacturers show higher CO■ savings; and how manufacturing is distributed by country.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

A typical viewer opens the published dashboard, scans the overall layout, selects a visualization of interest, compares categories or manufacturers, interprets the chart labels and values, and uses the findings as a starting point for further questions. This journey is illustrative because no formal user-testing observations were provided.

3.2 Solution Requirement

The solution requires a prepared EV dataset, a visualization tool, meaningful measures and dimensions, suitable chart types, clear titles and axes, a dashboard layout that groups related views, and a public link for access. The eight required views are listed in Section 7.

3.3 Data Flow Diagram

High-level data flow: Supplied EV dataset → Tableau data source → worksheet-level aggregations and visual encodings → eight visualizations → combined dashboard → Tableau Public publication. This is a textual representation of the demonstrated workflow; a formal DFD graphic was not supplied.

3.4 Technology Stack

Primary visualization platform: Tableau Public. Data input: the EV dataset supplied for the capstone. Repository: GitHub. The exact dataset filename, source URL, file format, record count and any additional preprocessing tools were not provided in the available project evidence.

4. PROJECT DESIGN

4.1 Problem–Solution Fit

A single dashboard is a practical way to consolidate multiple EV market indicators and reduce the effort required to compare them across separate tables or files. The dashboard provides the visual comparison layer; conclusions about user satisfaction or decision impact would require user testing.

4.2 Proposed Solution

The proposed solution is a Tableau dashboard containing eight coordinated views: Top 10 Manufacturers by Annual Sales; Annual EV Sales Trend; Battery Technology Distribution; Average Driving Range by Vehicle Category; Price vs Driving Range; Customer Ratings by Manufacturer; CO■ Savings by Manufacturer; and EV Manufacturing by Country.

4.3 Solution Architecture

The source dataset is connected to Tableau. Fields are assigned appropriate dimensions and measures, then used in separate worksheets. The worksheets are arranged in a dashboard and published to Tableau Public. No Snowflake database, warehouse, SQL pipeline or Streamlit application is evidenced in this project implementation.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work can be organized into the following sequence: understand the project brief and dataset; identify business questions; prepare the data and verify field types; create the eight worksheets; arrange and review the dashboard; publish to Tableau Public; prepare the GitHub repository; and record the project demonstration. Actual task dates and individual ownership were not supplied, so this sequence describes the workflow rather than a verified historical schedule.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The supplied dashboard screenshot confirms that eight visualizations are present. Formal load-time tests, dataset-size measurements, record counts, filter-efficiency comparisons, query histories, null/duplicate counts and performance benchmarks were not provided. These should be recorded from the actual workbook and source dataset rather than estimated.

7. RESULTS

7.1 Output Screenshots

The supplied dashboard screenshot shows all eight views. The visible chart structure supports the following types of analysis:

Visualization	What it communicates
Top 10 Manufacturers by Annual Sales	Compares annual sales units among leading manufacturers.
Annual EV Sales Trend	Shows annual sales movement across model years.
Battery Technology Distribution	Compares the representation of battery technology types.
Average Driving Range by Vehicle Category	Compares average driving range in kilometres across vehicle categories.
Price vs Driving Range	Shows the distribution of vehicle price against average driving range.
Customer Ratings by Manufacturer	Compares recorded customer-rating values across manufacturers.
CO■ Savings by Manufacturer	Compares recorded CO■ savings by manufacturer.
EV Manufacturing by Country	Maps the geographic distribution of manufacturing records.

Insert the original Tableau dashboard screenshot in this section when preparing the final submission copy. Published dashboard: https://public.tableau.com/app/profile/amushka.soni/viz/tableu_17931290491940/Dashboard2

8. ADVANTAGES & DISADVANTAGES

Advantages: combines eight views in one place; uses different chart types for different questions; supports manufacturer, category, technology and country comparisons; and is accessible through Tableau Public.

Limitations: analysis is limited to the supplied dataset; exact dataset provenance and refresh frequency are not documented here; no measured performance results were supplied; and interactive filters or a separate Tableau Story are not confirmed by the available screenshot.

9. CONCLUSION

The project delivers a Tableau Public dashboard that organizes eight EV market visualizations into one view. It provides a structured way to compare annual sales, manufacturers, battery technologies, driving range, price, customer ratings, CO■ savings and manufacturing geography. Further validation using the source dataset and user feedback would strengthen the conclusions drawn from the dashboard.

10. FUTURE SCOPE

Possible future work includes adding verified filters for model year, manufacturer, battery type and vehicle category; refreshing the dataset on a documented schedule; adding data-quality checks; measuring dashboard load time; improving accessibility and mobile layout; and evaluating whether a governed database or access-controlled analytics platform is needed for larger or sensitive datasets.

11. APPENDIX

Source Code / Workbook: Tableau workbook files are included in the project repository.
Dataset Link: exact public dataset URL not supplied; insert the original source link if available.
GitHub: https://github.com/anushkasoni/EV-market-analysis
Tableau Public Dashboard: https://public.tableau.com/app/profile/amushka.soni/viz/tableu_17931290491940/Dashboard2
Project Demo Link: insert the shareable recorded demo video URL after uploading it.

Submission note: Team ID intentionally left blank. Replace the blank with the assigned ID only if required. Add the actual dashboard screenshot, dataset citation, exact GitHub URL confirmation and demo video link before final submission.